EXAMPLE — what good looks like · illustrative reconstruction, grounded in a real cleared device

Security Architecture Views

Grounded in: BD Alaris™ Infusion System
Basis: FDA 2025 guidance — four views.

Device	BD Alaris™ Infusion System
Basis	FDA Premarket Cybersecurity Guidance (2025) §V.A.3 — four architecture views
Note	Illustrative reconstruction; not BD's confidential file.

1. Global system view

PCU + pump/syringe/PCA/respiratory modules ↔ facility network ↔ EMR interface ↔ device manager ↔ signed update service, with trust boundaries at each hop.

2. Multi-patient harm view

A compromised server / EMR interface must not push unsafe programs fleet-wide. Controls: signed/validated orders, server hardening, network segmentation, rate/range limits.

3. Updateability / patchability view

Signed software distributed to the installed base; integrity verified on receipt; anti-rollback; staged rollout with monitoring.

4. Security use-case views

Use case	Key controls
Clinician auth → program infusion	Authentication, authorization, audit log
EMR auto-program	Mutual auth, order validation, range limits
Remote service	Authenticated, least-privilege, logged
Software update	Signed image, integrity check, anti-rollback

Sources (public)

- FDA / BD — “BD Receives FDA 510(k) Clearance for Updated BD Alaris Infusion System” (Jul 21, 2023).
- FDA — Cybersecurity in Medical Devices: QMS Considerations and Content of Premarket Submissions (final, Jun 27, 2025).
- AAMI SW96; ISO 14971; IEC 81001-5-1.